

RECOMMENDING TESTS IN A MULTIMEMBERSHIP BAYESIAN DIAGNOSTIC EXPERT SYSTEM

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Abstract

A computerized test selection module for a multimembership Bayesian diagnostic expert system has been built to recommend tests to further refine the differential diagnosis. The output of our program includes recommendations for individual tests, for groups of tests to be administered in sequence, and for groups of parallel tests. The original set of available tests is large. This generates a huge number of groups of tests, any of which could possibly provide the most valuable information. Selection of a proper subset of tests from the original set is critical.

1 Introduction

The use of laboratory tests has been a major factor in the increasing cost of medical care during past decades. These costs are largely due to an extraordinary increase in the type and volume of tests ordered and performed in hospitals [1]. New technology has produced many new tests. These tests increase our diagnostic ability, but many of them are extremely expensive. The use of additional tests may comfort the patient and enhance the physician's belief that all diagnostic avenues have been pursued, but more tests do not necessarily produce more certainty; on the contrary, some tests will cause higher risk to the patient without improving diagnostic certainty [2].

Decisions about what tests should be ordered are difficult for the physician. In ordering tests, the physician needs to consider the contribution that these tests will provide, as well as the costs (including risk to the patient, time required to obtain results, actual dollar costs, etc.) of these tests. Automated test selection can improve physicians' decision making and make them more comfortable about decisions already made.

The purpose of this test selection module is to suggest appropriate tests to the physician on the basis of decision analysis techniques.

2 Bayesian Diagnostic Model

MEDAS (the Medical Emergency Decision Assistant System) is an expert system that employs a multimembership Bayesian model as its inference engine that allows the system to recognize multiple disorders in the same patient [3]. Disorder patterns are composed of sets of features that are significant for diagnosis. The term feature refers to any piece of information related to the patient health status such as measurements, signs, symptoms, laboratory test results, etc. The Bayesian model requires that features have binary values, either positive (+) or negative (−).

The MEDAS system records feature interactions in order to ensure that the features recorded for an individual patient are as independent as possible. There are four basic types of features in MEDAS: independent binary features (Headache), mutually exclusive binary features (Sex is male or female), value features (White blood count), and compound features (Sex = female and Hematocrit > 42). Value features are transformed into a group of exclusive binary features by using value ranges. Exclusive features exist in groups. Only one of the features in each group can be positive; when one of the features is set positive the others are ignored. A compound feature which is constructed from several binary features is set only when all the component features are set.

Bayes' formula appears in several variations [4]. MEDAS applies Bayes' formula to the relationship of features and disorders as [3]:

$$P(D_i | F_1, \dots, F_k) = \frac{P(D_i) \cdot f_i(F_1) \cdots f_i(F_k)}{P(D_i) \cdot f_i(F_1) \cdots f_i(F_k) + (1 - P(D_i)) \cdot \bar{f}_i(F_1) \cdots \bar{f}_i(F_k)}$$

where $P(D_i)$ is the prior probability of disorder D_i and $P(D_i | F_1, \dots, F_k)$ is the

posterior probability for disorder D_i after observing the features $F_1, \dots, F_k$.

Let P_{ij} = the conditional probability that feature F_j is positive, given that disorder D_i is present.

Let $\bar{P}_{ij}$ = the conditional probability that feature F_j is positive, given that disorder D_i is absent.

If F_j is positive, then $f_i(F_j) = P_{ij}$ and $\bar{f}_i(F_j) = \bar{P}_{ij}$. If F_j is negative, then $f_i(F_j) = 1 - P_{ij}$ and $\bar{f}_i(F_j) = 1 - \bar{P}_{ij}$. The P_{ij} values represent the frequency of true positives, namely the sensitivity of feature F_j with respect to disorder D_i . The $\bar{P}_{ij}$ values represent the frequency of false positives, namely one minus the specificity (which indicates the frequency of negative test results in patients without the disorder) of feature F_j with respect to disorder D_i [3, 4].

3 Test Selection

The test selection module that we designed is based on the MEDAS system but it should work just as well in other Bayesian diagnostic expert systems. Our test selection module serves two primary purposes:

1. Recommending optimal individual tests or groups of tests to the physician during the diagnostic process. This module produces three types of output: recommended individual tests, recommended groups of tests to be performed in a specified sequence, and recommended groups of tests that can be performed simultaneously.
2. Identifying as many disorders as we can.

The test selection module is designed based on the following assumptions:

1. The disorder patterns in the MEDAS knowledge base contain reliable values for sensitivity (P_{ij}) and specificity ($1 - \bar{P}_{ij}$) for each feature. These sensitivity and specificity values are the basic elements in our algorithm for selecting proper individual tests or groups of tests that can provide valuable information.
2. The information about the patient's complaints, history, and physical examination is correctly recorded. Errors in any of this information may cause the system to generate an incorrect differential diagnosis before any test can be suggested and the tests recommended on the basis of this differential diagnosis may also be in error.
3. The patient has one or more disorders in the current knowledge base. Our algorithm is intended to identify all the patient's disorders even those that do not necessarily relate to patient's complaints, physical examination, or routine tests. There is always a possibility that a potential disorder exists but could not be observed at the time when the information was recorded.

The test selection proceeds step by step as shown in Figure 1.

1. History and physical examination features. The first step in the test selection process is to determine if there are any additional features in these categories that could affect the differential diagnosis. Since this process involves only asking the patient a question or redoing a part of the physical examination, it should be done first. Since the cost of collecting this kind of information is relatively small compared to laboratory tests, we gather as much useful information as we can at this step. This information can then be used to generate a new differential diagnosis.
2. Hypothesis setting. The diagnostic hypotheses are set in this step. The physician may either set up hypotheses for the system to use in test selection or ask the system to generate hypotheses. If the system is asked to generate hypotheses, it will use the differential diagnosis from step 1.
3. Laboratory test recommendations. In the third step, we use the hypotheses selected in step 2 to make laboratory test recommendations that will help the physician to decide which disorders the patient is most likely to have.

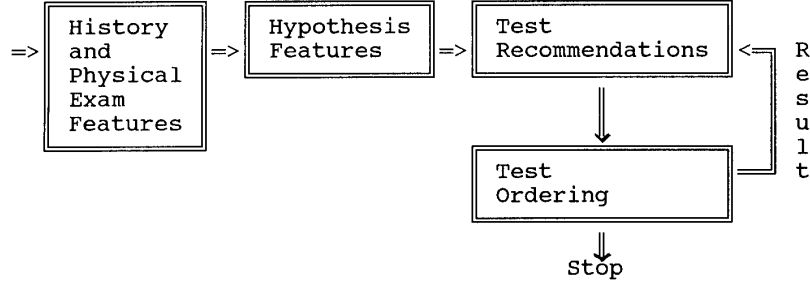


Figure 1. The Steps in the Test Selection Process.

Detailed discussions of these steps are given in the following sections.

4 History and Physical Examination Features

We attempt to collect information about the patient's history and physical examination before setting the diagnostic hypotheses. Most of this information is acquired by questions and answers. The response time of the system is the major concern here. Since the cost of collecting this information is relatively small compared to laboratory tests, we gather as much useful information as we can at this step. The information is gathered by asking questions one by one. The question is generated by considering the effect that each individual positive or negative feature has on the diagnosis. There are two steps in generating a question:

1. Building the set of relevant tests. The set of relevant features (SRF) is generated by forming the union of features in the history and physical examination categories appearing in the disorder patterns of disorders whose current posterior probabilities are greater than their original prior probabilities.
2. For all tests in the SRF, Shannon's Entropy [5]

$$H(F_j) = \sum_{i=1}^m E[h^i(P(D_i | F_j))]$$

where

$$h^i(P(D_i | F_j)) = -P(D_i | F_j) \log_2(P(D_i | F_j)) - (1 - P(D_i | F_j)) \log_2(1 - P(D_i | F_j))$$

is used to calculate the expected total weighted uncertainties of all these features. The features with minimum expected total weighted uncertainty are used to generate the question to be asked.

5 Hypothesis Setting

After history and physical examination information is collected, the diagnostic hypotheses are set either by the physician or by the system. At this step, the physician can set up the hypotheses in such a way that the tests recommended depend entirely on these hypotheses.

The diagnostic hypotheses generated by the system are determined as follows. First, we generate a list of disorders containing all those disorders whose posterior probabilities are higher than their original prior probabilities. The mean of the probabilities of the disorders in this list is calculated. If the number of disorders in the list is greater than some fixed N then the list of hypotheses includes those disorders with probabilities higher than $\text{mean} \cdot (3/2)$; otherwise, the hypotheses are set to those disorders with probabilities higher than the mean.

6 Laboratory Test Recommendations

The goal of test selection is not only to differentiate the disorders in the given list of hypotheses but also to verify multiple disorders (the patient could have more than a single disorder) from the given hypotheses. The set T of tests is created by forming the union of test features appearing in the disorder patterns of the disorders in the hypotheses.

For every test in the set T , there is an associated cost. The cost is a constant generated by a cost function considering the risk to the patient, the time required to obtain results, actual dollar costs, etc. The "cost efficiency" of a test is defined as the relationship between the cost and the expected total weighted uncertainty derived from Shannon's Entropy.

There are three different outputs of tests that our test selection module could recommend: individual tests, sequential tests, and parallel tests.

6.1 Individual Tests

The expected total weighted uncertainty is generated using Shannon's Entropy. The test whose value V (the ratio of the expected total weighted uncertainty to the cost) is the smallest will be the best selection. For instance, a test X with cost 5 units and expected total weighted uncertainty of 2 will be better than a test Y with cost 4 and an expected total weighted uncertainty of 4 units. For every test in T , we calculate the value V for each test. The B tests with the smallest V values will be recommended, where B is the maximum number of individual tests to be recommended.

6.2 Sequential Tests

A decision tree is used to determine the best groups of tests to recommend when the tests in the groups are tested sequentially. The depth D of the tree decides the number of tests in the group. There are two different results. The expected total weighted uncertainty is used in calculating the effect of each test carried out.

Total cost = $\sum_{i=1}^D$ Cost of test i). The cost efficiency is the expected total weighted uncertainty/total cost.

For instance, a group of two tests $\{X, Y\}$ leads us to consider eight possible outcomes:

$[X = +, Y = +]$ with probability $P(X = + | D) * P(Y = + | (D \wedge X = +))$.
 $[X = +, Y = -]$ with probability $P(X = + | D) * P(Y = - | (D \wedge X = +))$.
 $[X = -, Y = +]$ with probability $P(X = - | D) * P(Y = + | (D \wedge X = -))$.
 $[X = -, Y = -]$ with probability $P(X = - | D) * P(Y = - | (D \wedge X = -))$.
 $[Y = +, X = +]$ with probability $P(Y = + | D) * P(X = + | (D \wedge Y = +))$.
 $[Y = +, X = -]$ with probability $P(Y = + | D) * P(X = - | (D \wedge Y = +))$.
 $[Y = -, X = +]$ with probability $P(Y = - | D) * P(X = + | (D \wedge Y = -))$.
 $[Y = -, X = -]$ with probability $P(Y = - | D) * P(X = - | (D \wedge Y = -))$.

where

$P(X = + | D)$ is the probability that test X is positive given disorder D .

$P(Y = + | (D \wedge X = +))$ is the probability that test Y is positive given disorder D after X is tested positive.

In the sequential testing method, the result of testing X first and then testing Y is the same as the result of testing Y first and then testing X . The unnecessary calculations of different orders in a group can be avoided. There are problems: 1. how deep should the tree grow? 2. in which order that tests should be performed? The depth of the tree is set up depending on the response time that the user requires and the accuracy of the decision. The smaller the depth, the faster the response time and the smaller the degree of accuracy. The order of the tests in the group can be ranged according to their individual cost efficiency.

6.3 Parallel Tests

Performing tests in parallel is the same as considering any one of the tests to meet our expected result. For instance, a group contains two tests X and Y . The expected testing result is either X tested positive or Y tested negative. The probability that the expected result will occur is $1 - P(X \text{ is negative}) \cdot P(Y \text{ is positive})$.

The probability of occurrence will be much higher but the cost will be certainly higher. The parallel testing method can be used in an emergency case where there is no time to wait for the results of sequential tests to come back.

Selecting a best individual test among N tests is quite simple since any outcome that a single test could produce can be calculated easily. Most of the time, an individual test will not significantly achieve our goal. An appropriate group of tests needs to be considered. A group of tests can be performed sequentially or simultaneously. The two best individual tests do not necessarily contribute a best two-element sequence [6].

The number of i-element test groups given N possible tests [7] is

$$\binom{N}{i}$$

where any test in the groups has two possible results, either positive (+) or negative (-). Thus the total number of all possible outcomes generated by an i-element test group [7] is

$$\binom{N}{i} \times 2^i$$

If all possible test groups are considered, then the total number of all possible outcomes will be

$$\sum_{i=1}^N \binom{N}{i} \times 2^i$$

This is too large a number of possible test groups for consideration even by the computer. In order to reduce this number, either to select the M best tests of the original N tests or to generate a reasonable number of test sequences from the N original tests, we use special calculation techniques.

As long as P_{ij} and $\bar{P}_{ij}$ are nonzero we calculate every ratio

$$R_{ij} = \frac{P_{ij}}{\bar{P}_{ij}} \text{ if } P_{ij} \geq \bar{P}_{ij} \text{ and } R_{ij} = \frac{1 - \bar{P}_{ij}}{1 - P_{ij}} \text{ if } P_{ij} < \bar{P}_{ij}$$

for test j associated with every hypothesis i. Let D be the maximum depth of a decision tree and the maximum number of tests that need to be performed simultaneously. For each diagnostic hypotheses i, there is a new list i containing the D tests with the highest ratio R_{ij} in the original test list. The union of these new lists will be a new set of tests. The size of this new set of tests is smaller than the original set of tests (at most the same size).

7 Conclusion

Our test selection module has the ability both to increase the distance between diagnoses and the ability to identify multiple disorders. Both individual tests and groups of tests have been considered in our test selection module. Selecting a smaller size test subset which can produce the optimal result from the original test set is not an obvious exercise. The simplest but the most unreliable way of constructing a test set is to select any specific number D individual best tests from the original set [8]. We propose a more reasonable way to construct this proper subset. A smaller proper subset of tests will improve the system response time by avoiding unnecessary calculation.

References

- [1] Griner, P. F. and Liptzin, B., Use of the Laboratory in a Teaching Hospital: Implications for Patient Care, Education, and Hospital Costs, *Clinics of Internal Medicine*, Vol. 75, No. 2, Aug. 1971, 157–163.
- [2] Kassirer, J. P., Sounding Board – Our Stubborn Quest for Diagnostic Certainty: A Cause of Excessive Testing, *The New England Journal of Medicine*, Vol. 320, No. 22, Jun. 1989, 1489–1491.
- [3] Ben-Bassat, M., Carlson, R. W., Puri, V. K., Davenport, M. D., Schriver, J. A., Latif, M., Smith, R., Portigal, L. D., Lipnick, E. H. and Weil, M. H., Pattern-Based Interactive Diagnosis of Multiple Disorders: The MEDAS System, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. PAMI-2, No. 2, Mar. 1980, 148–160.
- [4] Galen, R. S., Application of the Predictive Value Model in the Analysis of Test Effectiveness, *Clinics in Laboratory Medicine*, Vol. 2, No. 4, Dec. 1982, 685–699.
- [5] Ben-Bassat, M., Multimembership and Multiperspective Classification: Introduction, Applications, and a Bayesian Model, *IEEE Transactions on Systems, Man, and Cybernetics*, Vol. SMC-10, No. 6, Jun. 1980, 331–336.
- [6] Cover, T. M., The Best Two Independent Measurements Are Not the Two Best, *IEEE Transactions on Systems, Man, and Cybernetics*, Jan. 1974, 116–117.
- [7] Szolovits, P. and Pauker, S. G., Categorical and Probabilistic Reasoning in Medical Diagnosis, *Artificial Intelligence* 11, 1978, 115–144.
- [8] Devijver, P. A. and Kittler, J. *Pattern Recognition: A Statistical Approach*, Prentice-Hall Inc., Englewood Cliffs, NJ, 1982.